Comprehensive Overview of Machine Learning and Logistic Regression

1. Introduction to Machine Learning

Machine Learning (ML) is a subfield ofartificial intelligence focused on building systems that learn from data to improve their performance on a specific task without being explicitly programmed. It involves the use of algorithms to identify patterns in data and make decisions or predictions.

2. Major Types of Machine Learning

Type	Definition	Typical Tasks
Supervised Learning	Learning from labelled data where the model is provided with input-output pairs.	Classification, Regression
Unsupervised Learning	Learning from unlabelled data to find hidden structures or patterns.	Clustering, Association
Reinforcement Learning	Learning through trial and error by interacting with an environment to maximize rewards.	Game AI, Robotics

3. Logistic Regression Concepts

Despiteits name, Logistic Regression is a**classification** algorithm. It is used to estimate the probability that an instance belongs to a particular class (e.g., Yes/No, 0/1).

The Sigmoid Function

Logistic regression uses the Sigmoid function (also called the Logistic function) to map any real-valued number into a value between 0 and 1.

```
f(z) = 1 / (1 + e^-z)
```

Where z is the linear combination of inputs ($z = \beta_0 + \beta_1 x_1 + ...$).

Decision Boundary

A threshold (commonly 0.5) is applied to the probability to determine the final class. If the probability is greater than 0.5, the model predicts the positive class; otherwise, it predicts the negative class.

4. Evaluation Metrics

- **Accuracy:** The ratio of correctly predicted observations to the total observations.
- **Confusion Matrix:** A table used to describe the performance of a classification model, showing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

CODE

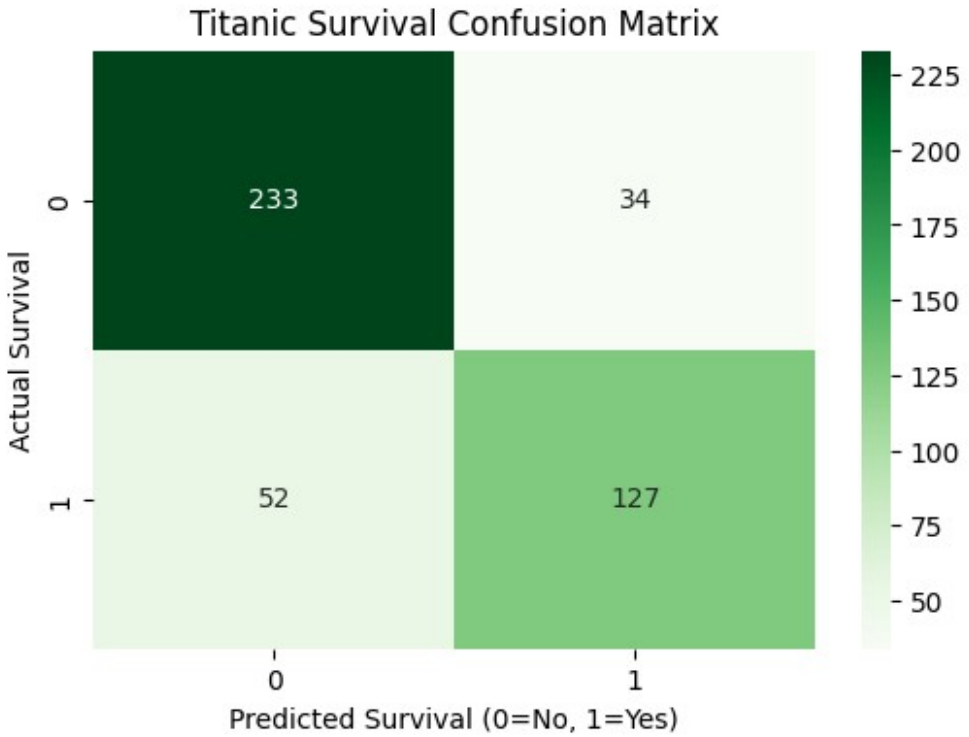
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
df = pd.read_csv('titanic.csv')
df.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'], axis=1, inplace=True) df['Age'] =
df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)

X = df.drop('Survived', axis=1)
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.5, random_state=42)
model = LogisticRegression(max_iter=500)
model.fit(X_train, y_train)
predictions = model.predict(X_test)
acc = accuracy_score(y_test, predictions)
print(f"Model Accuracy: {acc * 100:.2f}%")

cm = confusion_matrix(y_test, predictions)
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens')
plt.title('Titanic Survival Confusion Matrix')
plt.xlabel('Predicted Survival (0=No, 1=Yes)')
plt.ylabel('Actual Survival')
plt.show()


print("\nDetailed Performance Report:")
print(classification_report(y_test, predictions))
```

Model Accuracy: 80.72%



Detailed Performance Report:

	precision	recall	f1-score	support
0	0.82	0.87	0.84	267
1	0.79	0.71	0.75	179
accuracy			0.81	446
macro avg	0.80	0.79	0.80	446
weighted avg	0.81	0.81	0.81	446